

Topics Covered:

- Rotations
- Transformations
- The Adjoint Operation
- Body and Spatial Velocity
- Practice Problems

Additional Reading:

- MLS Chapter 2, LP Chapter 3

Rotations

In Euclidean space, the 3x3 rotation matrix that describes the orientation of frame B with respect to frame A is given by:

$$R_{ab} = [x_{ab}, y_{ab}, z_{ab}]$$

Key properties of rotation matrices include:

$$\begin{aligned} \det R &= +1 && \text{(Assuming the right hand rule)} \\ RR^\top &= R^\top R = I && \text{(And thus, } R^\top = R^{-1}) \end{aligned}$$

For rotations in the plane (about the z-axis that comes out of the page), we use the rotation matrix:

$$R(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

This generalizes to the canonical rotation matrices about the x , y , and z axes in 3D:

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix}, \quad R_y(\theta) = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix}, \quad R_z(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The set of all 3x3 matrices which satisfy the properties of rotation matrices is called the Special Orthogonal Group in 3D, denoted $SO(3)$. This group satisfies the following group axioms:

1. Closure: If $R_1, R_2 \in SO(3)$, then $R_1 R_2 \in SO(3)$
2. Associativity: $(R_1 R_2) R_3 = R_1 (R_2 R_3)$
3. Identity: The identity matrix is $I \in SO(3)$
4. Inverse: The inverse of $R \in SO(3)$ is $R^\top \in SO(3)$

Rotation matrices can be used to 1) Rotate Points, 2) Rotate Vectors, 3) Rotate Frames:

$$\begin{array}{ll}
 p_a = R_{ab} p_b & (p_b = (x_b, y_b, z_b) \text{ is a point relative to frame } B) \\
 v_a = R_{ab} v_b & (v_b = q_b - p_a \text{ is some vector in frame } B) \\
 R_{ac} = R_{ab} R_{bc} & (R_{ac} \text{ is the orientation of frame } C \text{ relative to frame } A)
 \end{array}$$

Transformations

General rigid body motion involves both translation and rotation. Thus, we describe the configuration of a rigid-body with a body frame B , relative to some frame A , as the pair (d_{ab}, R_{ab}) . This configuration space is called the Special Euclidean Group in 3D, denoted $SE(3)$:

$$SE(3) = \{(d, R) \mid d \in \mathbb{R}^3, R \in SO(3)\} = \mathbb{R}^3 \times SO(3)$$

There are two ways of talking about configurations: vector form and homogeneous form. In vector form, we describe the configuration as a pair (d_{ab}, R_{ab}) where $d_{ab} \in \mathbb{R}^3$ is the translation vector and $R_{ab} \in SO(3)$ is the rotation matrix. In homogeneous form, we describe the configuration as a 4x4 matrix:

$$g_{ab} = \begin{bmatrix} R_{ab} & d_{ab} \\ 0 & 1 \end{bmatrix}$$

These transformations can be used to rigidly transform a point (p) using the expressions:

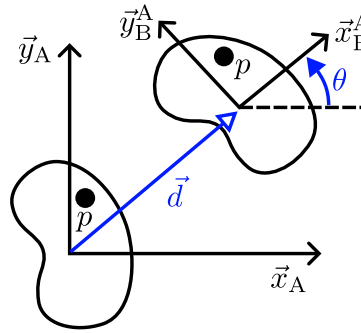
$$p' = \vec{d} + R_{ab}(\theta)p$$

or

$$p' = \begin{bmatrix} R & \vec{d} \\ 0 & 1 \end{bmatrix} p$$

These transformations can also be interpreted as representing the point p in frame A instead of on the local body frame.

This transformation is illustrated in the following diagram:



If we wanted to combine two different transformations together, there are also two ways of doing this for both the vector-form and homogeneous-form coordinates:

$$(d_{ac}, R_{ac}) = (d_{ab}, R_{ab})(d_{bc}, R_{bc}) = (d_{ab} + R_{ab}d_{bc}, R_{ab}R_{bc})$$

or

$$g_{ac} = g_{ab}g_{bc} = \begin{bmatrix} R_{ab}R_{bc} & d_{ab} + R_{ab}d_{bc} \\ 0 & 1 \end{bmatrix}$$

We can use this combination of transformations to describe the configuration of a robot manipulator. For example, let's consider the following manipulators:

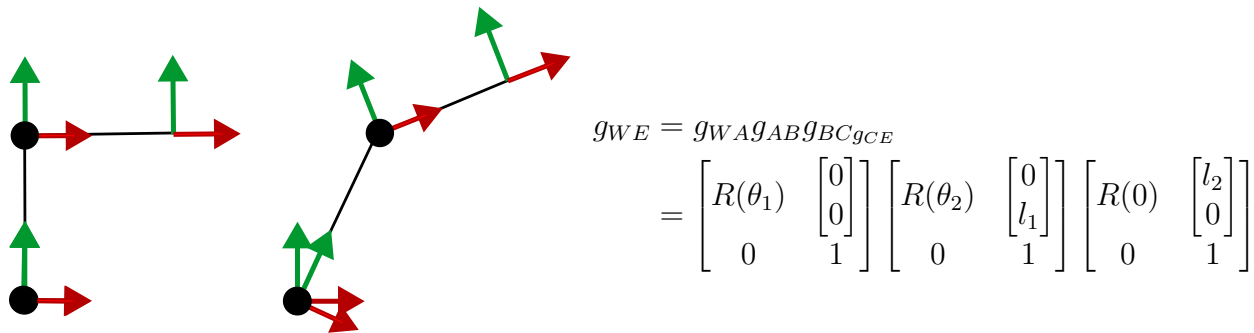


Figure 1: Example Manipulator 1: $\vec{\theta} = (\theta_1, \theta_2)$ with lengths l_1 and l_2

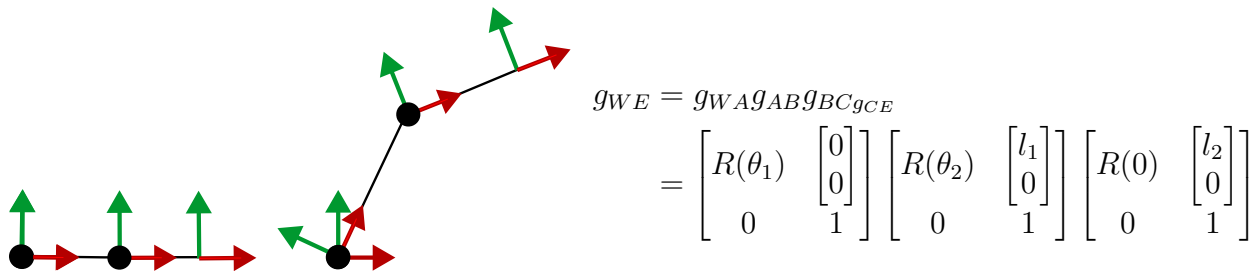


Figure 2: Example Manipulator 2: $\vec{\theta} = (\theta_1, \theta_2)$ with lengths l_1 and l_2

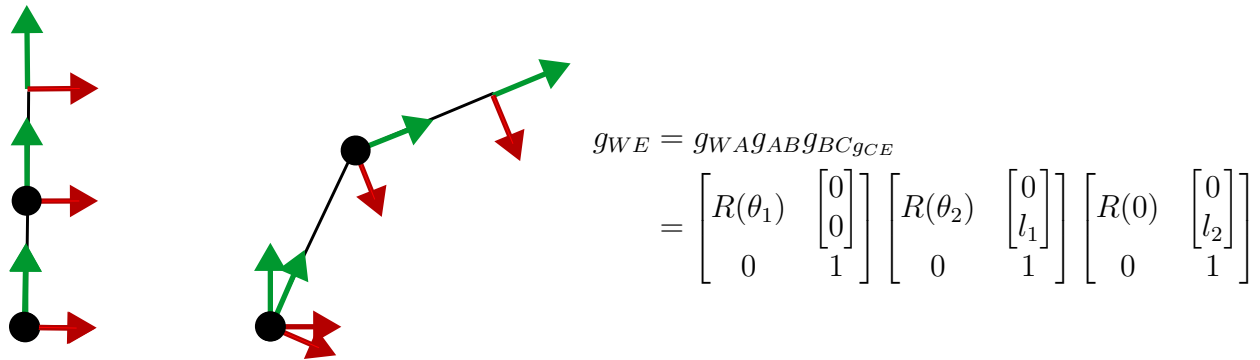


Figure 3: Example Manipulator 3: $\vec{\theta} = (\theta_1, \theta_2)$ with lengths l_1 and l_2

For all of these examples, note that you could derive the correct $\vec{d}$ vectors based on just the right-hand images since the frame axes are provided for you. But I've also provided you with the zero configurations on the left for clarity.

If we have robots with prismatic joints, this process is the same, we just have a variable in our $\vec{d}$ vector instead of a constant. For example, consider the following manipulator:

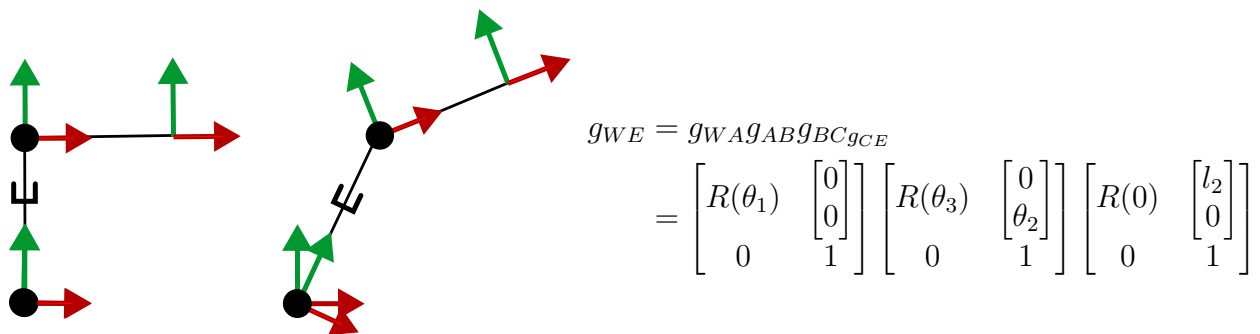


Figure 4: Example Manipulator 4: $\vec{\theta} = (\theta_1, \theta_2, \theta_3)$ where θ_2 is now the prismatic joint.

The Adjoint Operation

In general, this adjoint operation is used to change the frame of reference. Our classic example is if we wanted to change some transformation that was in the end-effector frame to a transformation of the tool frame. Specifically, assume that we have the transformations g_{we} and $g_{we'}$. We can use these to compute the transformation that the end-effector undergoes by:

$$g_{ee'} = g_{we}^{-1}g_{we'}$$

If we also have the transformation between the end-effector and some tool in the end-effector's grip, g_{et} , then we can use the adjoint to obtain the transformation of the tool:

$$g_{tt'} = g_{et}^{-1}g_{ee'}g_{et} = \text{Ad}_{g_{et}^{-1}}g_{ee'}$$

Always pay attention to the form of your adjoint operation. If you are given a different coordinate frame relationship (such as g_{te}), then your inverse will be in the opposite order:

$$g_{tt'} = g_{te}g_{ee'}(g_{te})^{-1} = \text{Ad}_{g_{te}}g_{ee'}$$

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Body and Spatial Velocity

The next topic we still need to cover is the concept of body and spatial velocity. In words, the body velocity is the velocity of the origin of the body coordinate frame relative to the spatial frame, as viewed in the current body frame. The spatial velocity is the velocity of a (possibly imaginary) point on the rigid body which is traveling through the origin of the spatial frame. The mathematical equations for these quantities are derived using the body and spatial twists:

$$\begin{aligned}\hat{\xi}_s &= \dot{g}_{sb}g_{sb}^{-1} \\ \hat{\xi}_b &= g_{sb}^{-1}\dot{g}_{sb}\end{aligned}$$

Which give us the expressions:

$$\begin{aligned}\hat{\xi}_s &= \begin{bmatrix} \dot{R}R^\top & -\dot{R}R^\top d + \dot{d} \\ 0 & 0 \end{bmatrix} \\ \hat{\xi}_b &= \begin{bmatrix} R^\top \dot{R} & R^\top \dot{d} \\ 0 & 0 \end{bmatrix}\end{aligned}$$

We can relate these two twists by the adjoint operation:

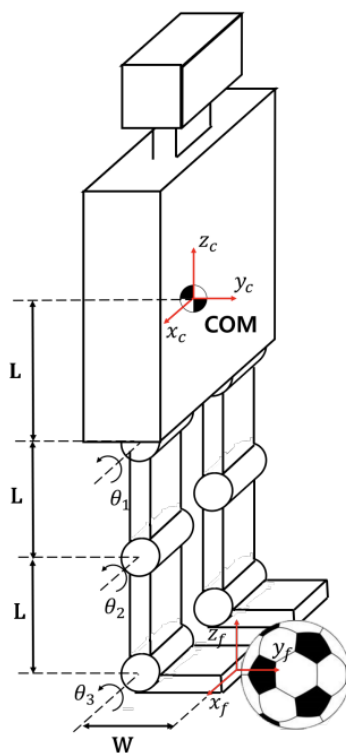
$$\begin{aligned}\hat{\xi}_s &= \text{Ad}_{g_{sb}} \hat{\xi}_b \\ &= g_{sb} \hat{\xi}_b g_{sb}^{-1}\end{aligned}$$

Practice Problems

The two problems below are the worked practice problems handed out ahead of the midterm. Try each one on your own first, then check against the solution that follows it.

Problem 1: Soccer Robot

The figure below shows a soccer robot in its zero configuration. The dimensions and joint variables are as indicated in the figure with $L = 1$ and $W = 0.5$.



- (a) Compute the forward kinematics expression g_f for the frame located at the tip of the foot with respect to the hip joint (a fixed frame located at the joint θ_1).
- (b) Assuming the robot's kicking leg is at the initial configuration $\vec{\theta} = (\theta_1, \theta_2, \theta_3) = (-\pi/2, -\pi/2, 0)$, what is the distance between the hip joint and the foot tip frame?

- (c) Assume that the hip joint is actuated so that an angular velocity is created of $\omega = (1, 0, 0)$. All other joint angles are held constant. What is the linear velocity of the foot tip frame?

Solution

We can use planar transformations along the y-z plane to simplify this problem. Note that you could also use 3D transformations, specifying that translations happen in the y-z plane and rotations happen about the x-axis. The solution is the same either way.

(a) The expression for our forward kinematics, with respect to a fixed frame at the hip joint, is:

$$g_f = \begin{bmatrix} R(\theta_1) & \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R(\theta_2) & \begin{bmatrix} 0 \\ -L \end{bmatrix} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R(\theta_3) & \begin{bmatrix} 0 \\ -L \end{bmatrix} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} I & \begin{bmatrix} W \\ 0 \end{bmatrix} \\ 0 & 1 \end{bmatrix}$$

(b) Plugging in the values for $\vec{\theta}$ we get:

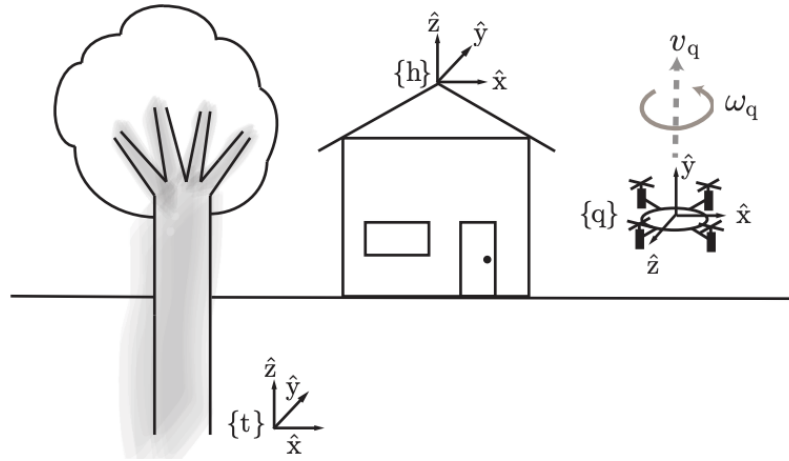
$$g_f = (-1.5, 1, -\pi)$$

(c) We can compute the linear velocity of the foot tip frame using the equation:

$$\begin{aligned} v &= \omega \times \begin{bmatrix} -1.5 \\ 1 \end{bmatrix} \\ &= [\omega]_{\times} \begin{bmatrix} -1.5 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1.5 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ -1.5 \end{bmatrix} \end{aligned}$$

Problem 2: Quadcopter, Tree, and House

Consider the scene below.



The scene depicts a quadcopter (frame q) flying near a tree (frame t) and house (frame h). The quadcopter is at a position with a displacement $d_{tq} = (10, 5, 5)$ m expressed in the tree frame t , and the house is at a position with a displacement $d_{th} = (0, 10, 10)$ m expressed in the tree frame t . The quadcopter is flying upwards with a velocity of 1 m/s, and rotating with a velocity of 1 rad/s (about the y -axis).

- (a) Calculate the quadcopter's twist in frame q and frame t .
- (b) Use the adjoint map to express the twist in the house frame h .

Provided below are some useful equations.

$$\begin{aligned} v_b &= R_{ab}^\top \dot{d}_{ab} & v_s &= -\dot{R}_{ab} R_{ab}^\top d_{ab} + \dot{d}_{ab} \\ \omega_b &= (R_{ab}^\top \dot{R}_{ab})^\vee & \omega_s &= (\dot{R}_{ab} R_{ab}^\top)^\vee \end{aligned}$$

Solution

(a) First, from the drawing we can obtain our rotation matrix as

$$\begin{aligned} R_{tq} &= R_x(\pi/2)R_y(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\pi/2) & -\sin(\pi/2) \\ 0 & \sin(\pi/2) & \cos(\pi/2) \end{bmatrix} \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \\ &= \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ \sin(\theta) & 0 & -\cos(\theta) \\ 0 & 1 & 0 \end{bmatrix} \end{aligned}$$

Thus, taking the derivative of this rotation matrix yields:

$$\dot{R}_{tq} = \begin{bmatrix} -\sin(\theta) & 0 & \cos(\theta) \\ \cos(\theta) & 0 & \sin(\theta) \\ 0 & 0 & 0 \end{bmatrix} \dot{\theta}$$

At the moment we're evaluating the velocity, $\theta = 0$ and $\dot{\theta} = 1$, giving us the final expression for $\dot{R}_{tq}$ as:

$$\dot{R}_{tq} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Lastly, from the information provided in the question, we know that $\dot{d}_{tq} = (0, 0, 1)$ m/s. Then, plugging in our expressions to the provided equations, we get:

$$\begin{aligned} v_b \text{ (quad frame)} &= R_{tq}^\top \dot{d}_{tq} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}^\top \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \\ \omega_b &= (R_{tq}^\top \dot{R}_{tq})^\vee \\ &= \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right)^\vee \\ &= \left(\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} \right)^\vee = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \end{aligned}$$

And for the tree (spatial) frame:

$$\begin{aligned}
 v_s &= -\dot{R}_{tq} R_{tq}^\top d_{tq} + \dot{d}_{tq} \\
 &= \begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 10 \\ 5 \\ 5 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 10 \\ 5 \\ 5 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} 5 \\ -10 \\ 1 \end{bmatrix} \\
 \omega_s &= (\dot{R}_{tq} R_{tq}^\top)^\vee \\
 &= \left(\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \right)^\vee \\
 &= \left(\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right)^\vee = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}
 \end{aligned}$$

Thus, the quadcopter's twist in body and spatial frame are:

$$\begin{aligned}
 \xi_b &= (v_b, \omega_b)^\top = (0, 1, 0, 0, 1, 0)^\top \\
 \xi_s &= (v_s, \omega_s)^\top = (5, -10, 1, 0, 0, 1)^\top
 \end{aligned}$$

(b) We can use the adjoint map to express the twist in the house frame h as:

$$\begin{aligned}
 \xi_h &= g_{th}^{-1} \cdot \xi_t \cdot g_{th} \\
 &= \begin{bmatrix} I^\top & -I^\top d_{hq} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} [\omega_t]_\times & v_t \\ 0 & 0 \end{bmatrix} \begin{bmatrix} I & d_{hq} \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -10 \\ 0 & 0 & 1 & -10 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 & 5 \\ 1 & 0 & 0 & -10 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 1 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & -1 & 0 & 5 \\ 1 & 0 & 0 & -10 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 1 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & -1 & 0 & -5 \\ 1 & 0 & 0 & -10 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

This leaves us with the adjoint in the house frame as:

$$\xi_h = (v_h, \omega_h)^\top = (-5, -10, 1, 0, 0, 1)^\top$$